Predictive Modeling and Risk Scoring for Bank Customer Churn

1. Project Overview

The objective of this project is to develop a machine-learning model that predicts the likelihood of bank customers leaving the bank. The analysis helps identify customers who may be at risk of churn so that banks can plan suitable customer-retention strategies.

2. Project Objectives

- Analyze customer information and identify churn patterns.
- Perform data cleaning and exploratory data analysis.
- Engineer additional features to improve predictive modeling.
- Develop machine-learning models to predict customer churn.
- Identify important features associated with churn.
- Generate customer-level churn probabilities and risk categories.

3. Tools and Technologies

- Microsoft Excel – Data cleaning, analysis, PivotTables and charts.
- SQL – Data analysis and query practice.
- Python – Machine-learning implementation.
- Pandas and NumPy – Data manipulation.
- Matplotlib – Data visualization.
- Scikit-learn – Model development and evaluation.
- Google Colab – Python development environment.
- Power BI – Dashboard and business insights.

4. Dataset Description

The project uses a bank customer dataset containing 10,000 records.

The target variable is Exited:

- 0 = Customer retained.
- 1 = Customer churned.

The dataset initially contained 2,037 churned customers and 7,963 retained customers, representing a churn rate of 20.37%.

5. Exploratory Data Analysis

The analysis identified several patterns:

- Germany had a higher churn rate (32.44%) than France (16.15%) and Spain (16.67%).

- Inactive customers had a higher churn rate (26.85%) than active customers (14.27%).
- The average age of churned customers was approximately 44.84 years, compared with 37.41 years for retained customers.
- Customers with one product had a churn rate of 27.71%.

These findings helped identify patterns for further predictive analysis.

6. Feature Engineering

Four additional features were created:

1. Balance_Salary_Ratio
2. Product_Density
3. Engagement_Product
4. Age_Tenure_Interaction

These features represent relationships between existing customer variables and were included in the predictive modeling process.

7. Machine-Learning Models

Two classification algorithms were developed:

- Logistic Regression
- Random Forest Classifier

The dataset was divided into:

- 80% training data.
- 20% testing data.

The data was preprocessed using One-Hot Encoding for categorical variables and StandardScaler for numerical variables.

8. Model Performance Comparison

Evaluation Metric	Logistic Regression	Random Forest
Accuracy	84.20%	83.85%
Precision	71.36%	59.29%
Recall	37.35%	65.85%
F1 Score	49.03%	62.40%
ROC-AUC	83.63%	86.16%

Random Forest achieved better recall, F1 score and ROC-AUC. It identified 268 churned customers out of 407 actual churn cases in the test dataset.

Therefore, Random Forest was selected as the leading model for this project's churn-risk analysis.

9. Feature Importance Analysis

The five most important features identified by the Random Forest model were:

1. Age – 25.68%
2. Number of Products – 14.10%
3. Balance – 8.37%
4. Balance_Salary_Ratio – 6.97%
5. Age_Tenure_Interaction – 6.74%

These scores indicate the features the model relied on most when making predictions. They do not establish causal relationships.

10. Customer Risk Scoring

The trained Random Forest model was used to generate churn probabilities for the complete dataset.

Customers were grouped into three categories using initial project thresholds:

- Low Risk: Below 30%.
- Medium Risk: 30% to below 60%.
- High Risk: 60% or above.

A customer-level Excel report was generated containing predicted churn probabilities and risk categories.

11. Business Recommendations

Based on the analysis, the bank could:

- Prioritize reviewing customers classified as high risk.
- Improve engagement with inactive customers.
- Study customer needs across different age groups.
- Review product usage and develop relevant service offerings.
- Monitor churn-risk predictions and evaluate retention campaign outcomes.

12. Limitations

- Model predictions are not guarantees of future customer behavior.

- Risk thresholds are initial project choices and require business validation.
- Feature importance does not prove causation.
- Predicted probabilities should be checked for calibration before being interpreted as precise real-world probabilities.
- Further testing and model improvements may increase performance.

13. Conclusion

This project demonstrates how data analysis and machine learning can support customer churn prediction in the banking sector.

Two classification models were developed and evaluated. Random Forest achieved an ROC-AUC of 86.16% and recall of 65.85%, making it the stronger model for identifying churn cases in this evaluation.

The final customer risk report provides a starting point for identifying customers who may need additional engagement and retention support.

The project demonstrates practical applications of Excel, Python, data visualization and machine learning in financial analytics.